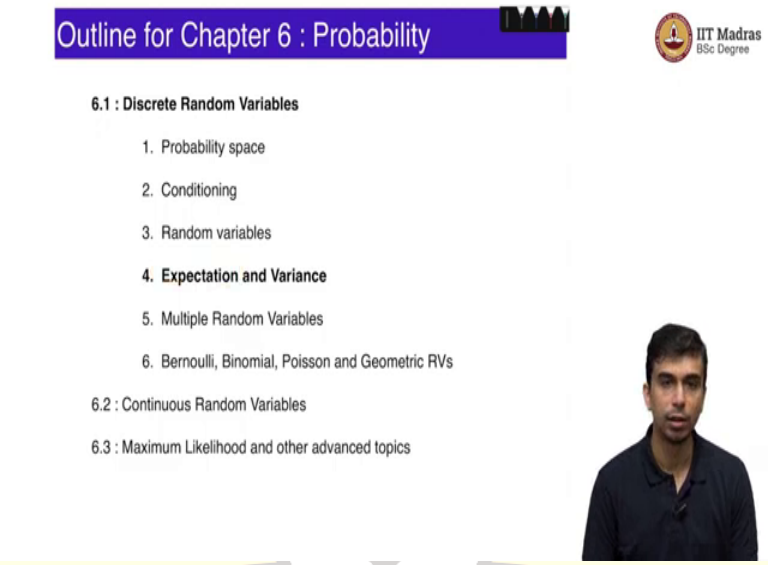


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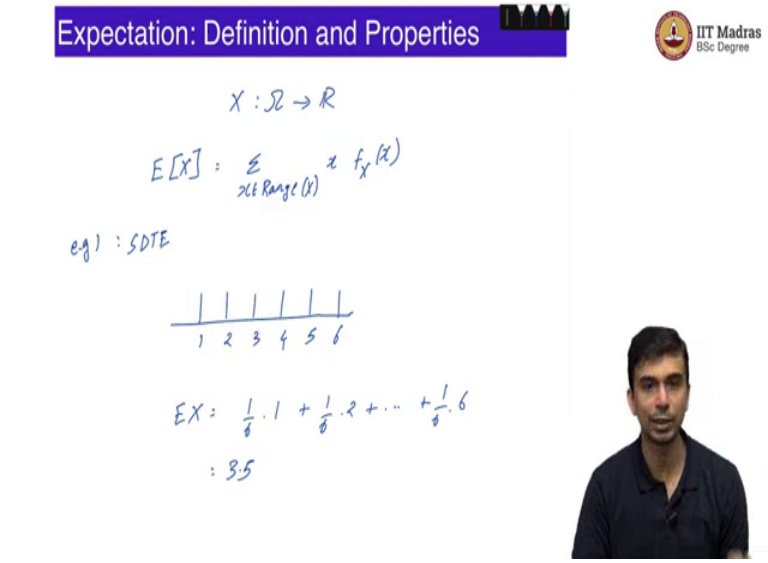
Machine Learning Foundations
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Expectation: definition and properties

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The next property, the next topic that we are going to look at is expectation and variance of random variables. These are, expectation is a measure of central tendency of a distribution and variance is a measure of its spread. We will make both of this clearer in today's, in this lecture.

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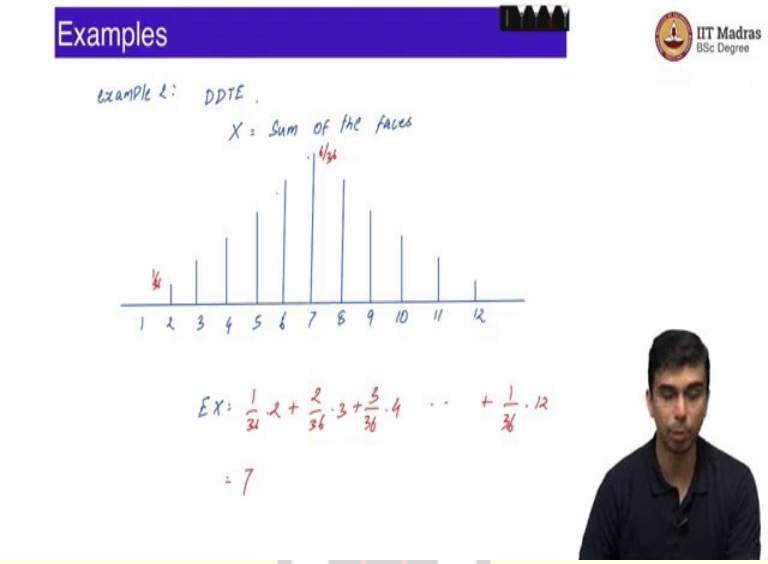
The expectation is a basic property of a random variable and it is defined completely in terms of its probability mass function. What is the, let us say, you have a random variable x, which is mapping from omega to R, and you have the experiment given by omega fp. The expectation of a random variable x, which is typically denoted by E of x, sometimes you might see the square brackets missing, but that is the typical notation. You will find expectation of square bracket of x is simply equal to sum over x in the range of the random variable x, x times fx of x. So, this is the definition of the expectation of a random variable.

What is an intuitive meaning for this? The intuitive meaning for this is that you can visualize the probability mass function and view the probability masses as literal masses. If that is the case, then the center of mass of the scale would effectively be the expectation. For example, let us take the example of the double, single dice throw experiment. Let us take a single dice throw experiment, where your x is simply the result of the first dice, result of the dice. So, in this case, what is the probability mass function?

Well, the probability mass function would take values between, the random variable would take value in 1, 2, 3, 4, 5, and 6, and the probability mass function would be 1 by 6 the 6 points. You can imagine the horizontal line as a scale and the masses as literal masses then the point at which you will balance this would be 3.5.

And we can, we will see that that is exactly what the expectation of x would be. In this case, expectation of x would be 1 by 6 times 1 plus 1 by 6 times 2 plus dot, dot, dot 1 by 6 times 6. This is exactly equal to 3.5. This is the definition of a center of mass as you might have seen in your basic physics class. So, this is a measure of central tendency. If you had to put a center of this random variable that is where you put, 3.5 is where you put a center. That is the expectation.

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You will see some more examples. Let us take the double dice throw experiment. And for now let us take the standard counting probability measure where all six faces are equally likely and the probability of any event is simply cardinality of that event by 36. So, if you have x to be the sum of the two faces, what you can show is that this, the density, the probability mass function would look like this.

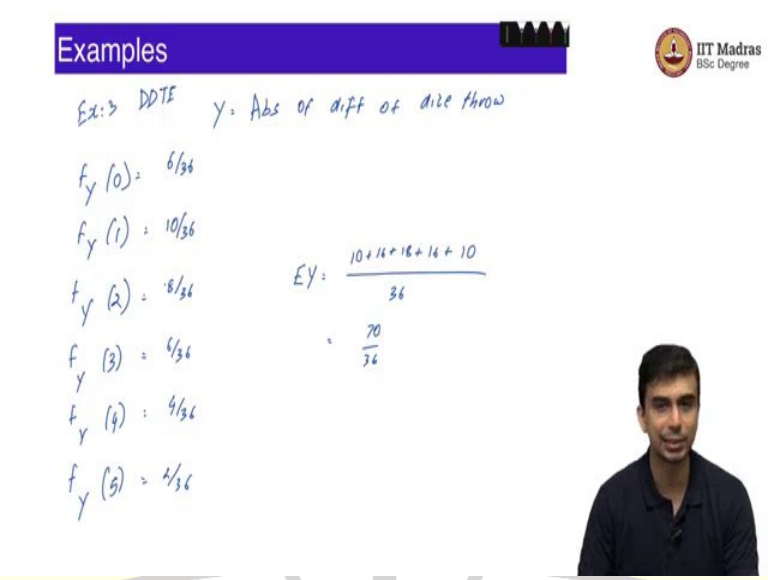
The probability of x is equal to 2 is 1 by 36. And probability of x equal to 3 is 2 by 36, 3 by 36 and so on. Probability of x is equal to 7 would be 6 by 36. And after 8 it will keep on decreasing to 5 by 36, 4 by 36, 3 by 36, 2 by 36, 1 by 36 and 0. These are, this is the probability mass function. We can easily see that the sum of all of these values would be exactly equal to 1. This is pretty easy to check from first principles.

For example, if you want to check what is the probability of the sum equal to 8, that simply you can note down the set of all elementary events that result in the sum being 8 that would be 2 comma 6, 3 comma 5, 4 comma 4, 5 comma 3 and 6 comma 2. There are six sort of five possible events here and 5 by 36 would be exactly this. And what is the center of, what is the expectation of this particular random variable from the graph you can easily guess that the center of this, this is symmetric around 7 and the center of mass of this will be 7 and in fact that is exactly the case.

You can compute the expectation of x from first principles here. If you did that, you would get 1 by 36 into 1 plus 2 by 36, sorry, 1 by 36 into 2 plus 2 by 36 into 3 plus 3 by 36 into 4 all the way

plus 1 by 36 into 12. If you did the summation you would get 7. And geometrically and graphically it also makes sense that the average or the central, if you throw two dice, the expected value is 7, because 7 is symmetric, the probability mass function of the sum is indeed symmetric around 7, and 7 is the largest, 7 is the most likely value also.

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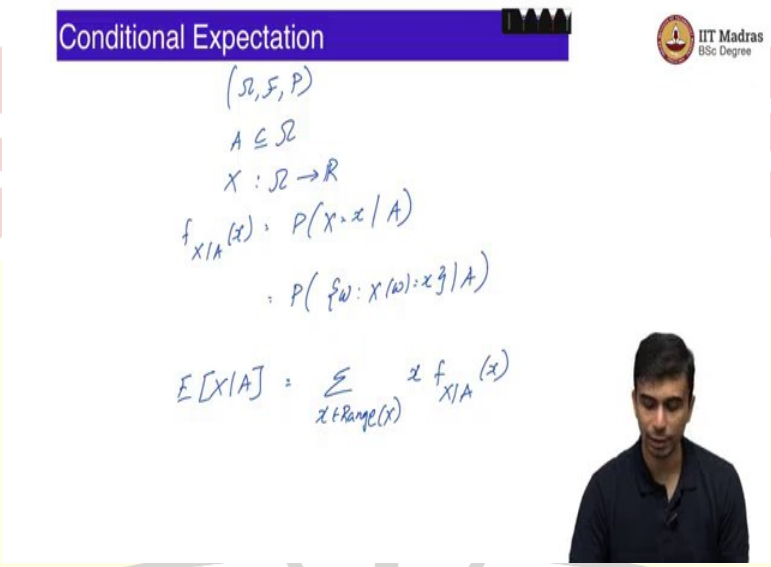


Let us do one more example. Let us say, take the same double dice throw experiment, let us say, y is the absolute value of the difference of the dice throw. So, if you have throwing two dice, let us say, it comes 4 comma 2, the difference, absolute difference is 2, absolute value of 4 minus 2 is 2. What is the probability of this, what is the probability mass function of this particular random variable? Well, fy of 0 is probability that both the dice, both the faces are equal, that is exactly equal to 6 by 36. What is the probability the difference is exactly equal to 1?

Well, you can note down all the events out of the 36 events that will be 10, you will count there are 10 possible events there, and 10 possible elementary events there. And well 10 possible elementary, elements are outcomes. So, the probability of that particular event is 10 by 36, and so on, you can compute the probability mass function. You can take values between 0 to 5. It cannot take values less than 0, because it is absolute value. And it cannot take values above 5, because the maximum way you can get the difference is 6 and 1. So, this would be the probability mass function of this particular random variable.

And you can compute the expected value of this random variable as using the simple formula, which is 6 by 36 into 0 plus 10 by 36 into 1 plus 8 by 36 into 2 plus 6 by 36 into 3 plus 4 by 36 into 4 plus 2 by 36 into 5. If you did that, you would get 70 by 36. The expected value of the difference is slightly less than 2. See these are just some simple examples to get you running.

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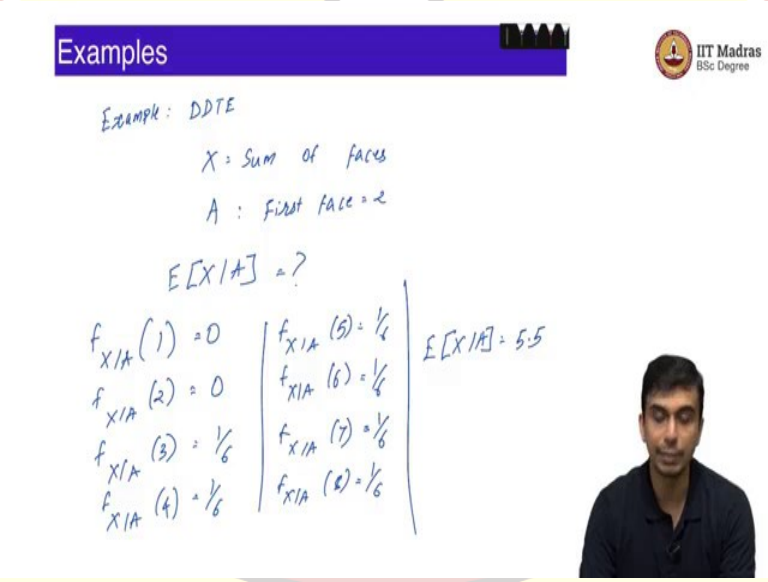
Now, we move on to the next concept which is conditional expectation. But before we go there, we must first define a conditional distribution. Once again, let us say, we are living in the probability experiment given by the tuple omega F, P. Let us say A is an event. A is a subset of omega. We, and x is some random variable. X is random variable from omega to R. We can define, I mean, we know what the probability mass function is. We can also define a conditional probability mass function as follows.

If X given A of x is probability of X is equal to x given A. So, if you did not have the conditional event here, it would simply be the standard PMF. But this is also a well-defined thing. So, probability x equal to x given A is the conditional probability. What is this? This is the probability of set of all omega such that x of omega equal to x given A. So, here we have the left hand side and right hand side are both valid events and conditional probabilities already defined. So, this is the conditional PMF of the random variable x given the event A. So, this is the conditional PMF.

And once you have the conditional PMF, you can define conditional expectation also. What is the expected value of x given the event A? What is that? Well, that is sum over x in range of x, x times fx given A of x. The only thing that has changed when compared to the standard expectation is that instead of using the PMF, we are using the conditional PMF. That is the only difference here.

A conditional explanation is a very powerful and useful tool and it makes a lot of reasoning. And most questions that you will be asking in practice would be the type of conditional expectation question. Let us have some simple examples.

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Let us do the example of the double dice throw experiment again. Let us say you want to know the expected value of the sum of the dice or let us say x is equal to the sum of faces that you see and let us say A is the event given by let us say first face is equal to 2. So, that is a valid event. You can ask the question, what is the expected value of x given A, what is that? That is the question. First off, you need to know the conditional PMF. What is that f X given A of, let us say, what is f X given A of 1?

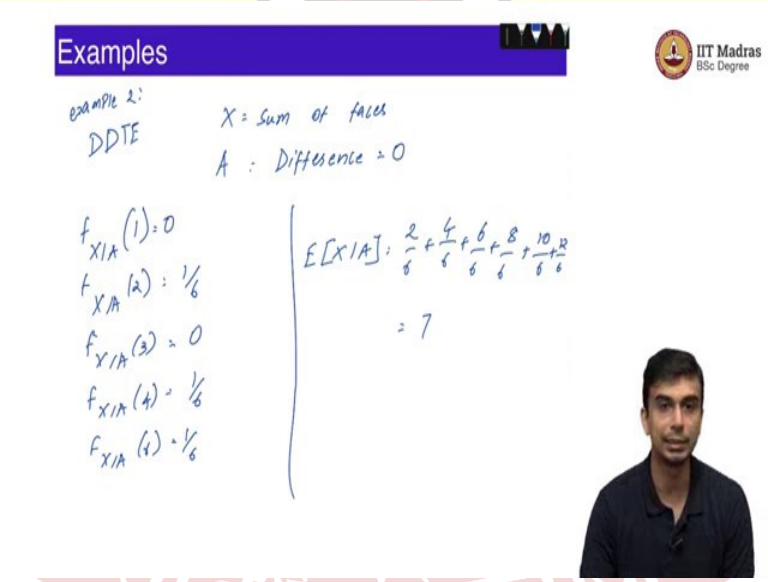
What is the event than the sum is equal to 1 given that the first face is equal to that is clearly 0. What is f of X given A of 2? What is the probability that the sum is 2 given that the first face is 2 that is also 0, because the smallest face is 1. The first face is already 2, that is equal to 0. What is

f X given A of 3? That is equal to 1 by 6. What is f X given A of 4? That is also 1 by 6. What is that?

What is f X given A of 4? What is the probability of the sum equal to 4 given that the first face is equal to 2? This is exactly equal to the probability that the second face is equal to 2, which is equal to 1 by 6. Similarly, you can compute this f X given A of 5 is also equal to 1 by 6, f X given A of 6 is also equal to 1 by 6, f X given A of 7 is also equal to 1 by 6, f X given A of 8 is also equal to 1 by 6. Everywhere else it is going to be 0.

You can, once you have the conditional PMF, you can compute the conditional expectation the same way, which would be expected value of X given A would simply be equal to 1 by 6 times 3 plus 1 by 6 times 4 plus 1 by 6 times 5 plus 1 by 6 times 6 plus 1 by 6 times 7 plus 1 by 6 times 8. That would exactly be equal to 5.5. Expected value of the sum of the faces given that the first face is equal to 2 is 5.5. This is a simple example.

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Let us do one more example. Let us say X is once again, the sum of, example, 2, let us say, sum of, X is the sum of faces and we are once again working in the double dice throw experiment. X is the sum of faces. And let us say A is the event, the difference equal to 0. Difference, it means that effectively both faces are the same. What is the, that is what A is.

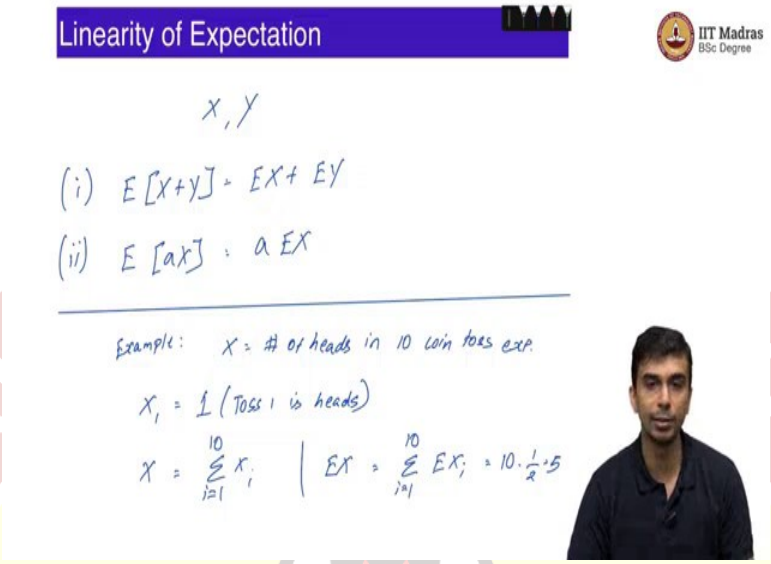
So, now, we can once again get the conditional PMF? What is f X given A of 1, which is 0, f X given A of 2. What is the probability that the sum of faces is 2 given that you know that the difference between the two faces is equal to 0? The answer is this is equal to 1 by 6 that only event this, I mean, given the difference is 0, the only six possibilities are 1 comma 1, 2 comma 2, 3 comma 3, 4 comma 4, 5 comma 5 and 6 comma 6.

So, f X given A of 2 is equal to 1 by 6. What is f X given A of 3? What is the probability that the sum is equal to 3 given that the difference is 0? The answer is 0. Because the 3 is an odd number, you can never get, if both the dice throws up the same, you can ever get an odd number, so it is 0.

Similarly, you can get f X given A of 4 is also equal to 1 by 6, f X given A of 6 is also equal to 1 by 6, f X given A of 8 equal to 1 by 6, f X given A of 10 equal to 1 by 6, f X given A of 12 is also equal to 1 by 6. Now, once you have this, you can compute the expected value of X given A would be equal to 2 into 1 by 6 plus 4 into 1 by 6 plus 6 into 1 by 6 plus 8 into 1 by 6 plus 10 into 1 by 6 plus 12 into 1 by 6.

That will be 2 by 6 plus 4 by 6 plus 6 by 6 plus 8 by 6 plus 10 by 6 plus 12 by 6, which is exactly equal to, sorry, I made a mistake here, is 8 by 6 plus 10 by 6 plus 12 by 6. What is the sum of this? Well, that is clearly equal to 7. The expected value of x given that, sum of faces given the difference is 0 is equal to 7. This is another example of a, simple example of computing conditional expectations.

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Now, we will move on to another property of expectation, which is going to be very useful. I am not going to prove this right now. But it is a very basic property. That is if you have multiple random variables, let us say, x and y, both are at the same probability experiment, of course, obviously, in this case, we can say expected value of x plus y is equal to Ex plus Ey. So, this is the first property, expected value of x plus y is equal to Ex plus Ey.

And then the second, let us say, a, some, you have some small scalar, let us say, you have a scalar a, what is expected value of ax? That is equal to a times expected value of x. So, these are two main properties of the expectation. This is, expectation is a linear operator. That is if you, let us say, the expected value of some random variable x is, let us say, 5. You can ask the question, what is the expected value of 2 times that random variable? That is equal to 2 times the expected value of that random variable which is equal to 10. That is what this statement is saying.

This is, and in fact, the more interesting statement is that expected value of x plus y equal to Ex plus Ey. So, you have that, irrespective of what, these x and y can be anything. They can be independent. We will not come to that yet. So, irrespective of what x and y are as long as they are both valid random variables, Ex plus y, E of x plus y is equal to Ex plus Ey. So, this is the property we will not prove right now, but this is something you should keep in mind. And this is very useful because it can help a lot in some examples which can be quite tough to get using the definition of expectation.

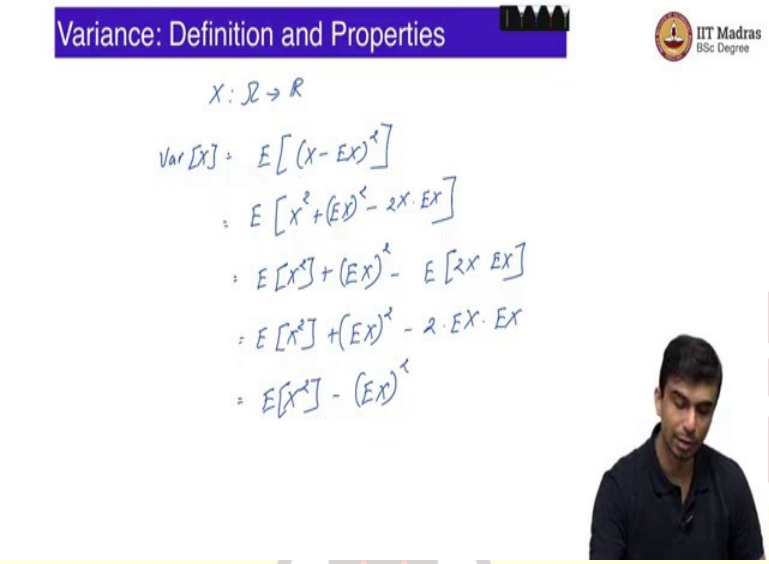
For example, let us say you have an experiment which is the 10 coin toss experiment. You have, you are tossing 10 coins, and the random variable X is number of heads. The experiment is a 10 coin toss experiment. And let us say you have a fair coin. So, effectively, this corresponds to a counting probability measure, where the probability of any event, sorry, probability of any event is simply the cardinality of that event divided by 2 power 10. So, the sample space of this particular experiment is 2 power 10. So, you have 10 coin tosses, so we can help H, H, H, H, H, H, H, all the way to T,T, T, T, T so that is 2 power 10 possible elements in the sample space.

Let us have the simplest thing. So, if X is the number of heads in the 10 coin toss experiment, you could, in principle, get the PMF of this random variable, probability of x equal to 0, probability of x equal to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 and then you could compute the expectation. I mean, once you have the PMF, you can compute the expectation. But here is the idea. Here you can use the linearity of expectation idea here. What you can do is you can define auxiliary random variables, let us say, x1 is equal to the indicator random variable of toss 1 is head.

Similarly, you can do x2, x3, x4, x5, x6, x7, x8, x9, x10, x is clearly sum over i equal to 1 to 10 xi. Clearly, you can see that the number of, total number of heads is simply the sum of these indicator random variables. So, now, Ex is by linearity of expectation simply the sum of the expectation of these things, sum over i equal to 1 to 10 EXi. But what is expected value of an indicator random variable? Expected value of an indicator random variable is particularly nice, because the expected value of indicator random variable is exactly equal to the probability of that random variable taking the value 1, which would exactly be equal to half, 10 into 1 by 2 which is equal to half.

So, now you immediately get that the expected number of heads in the 10 coin toss experiment is 5 without getting the probability mass function of this random variable and then doing this. So, this linearity of expectation is a pretty powerful tool that you will use often for computing explanations of more complicated random variables.

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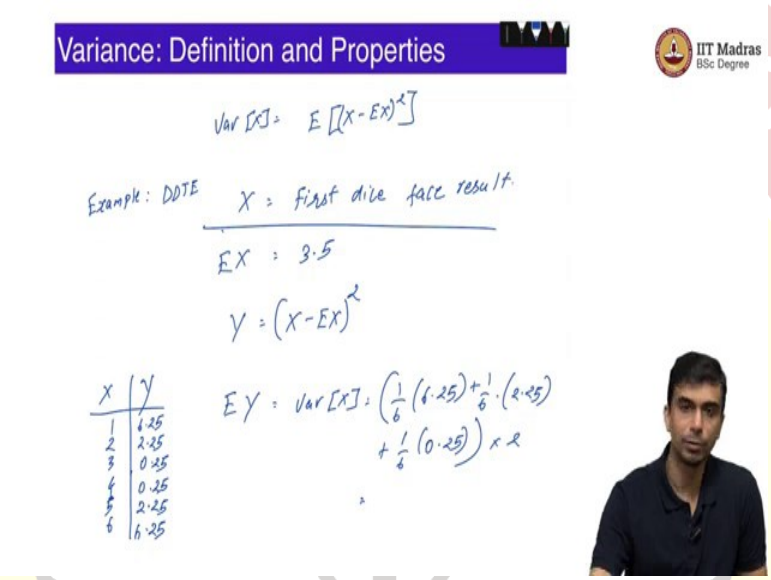
So, now, we will move to the, another property of a random variable, which is a variance. The variance is also just depends on the probability mass function in the same way that you had expectation. Variance, however, is different from expectation in the sense that variance is a measure of how much the random variable spreads out. Let us define the variance first. Let us say you have a random variable X from omega to R, the variance of the variable X is simply the expected value of the random variable given by X minus Ex the whole square.

Let us unpack this a little bit first. What is x? X minus Ex is a constant. So, once expectation of a random variable is a constant. So, x minus Ex is also, x minus Ex the whole square is a function of the random variable x and functions of random variables themselves are random variables. So, this entire thing is well defined. That is the first thing you should understand. So, expected value of x minus Ex the whole square let us expand a minus b the whole square just Ex square plus Ex the whole square minus 2x Ex. And here is where you can use the linearity of expectation.

So, you have expectation of sum of these three things. So, effectively, you can use a linearity of expectation to write, break this down further as expected value of x square plus Ex the whole square is a constant and expectation of a constant is simply equal to the same constant that is Ex the whole square minus 2x minus the expected value of 2x Ex. So, here is where you can use once again the linearity of expectation. Ex is a constant. So, you can take constants outside the expectation.

So, if you do that, you would get expected value of x square plus Ex the whole square minus, you can take the Ex outside and you can take the 2 also outside, so minus 2 times Ex into Ex, so which is exactly 2 times Ex the whole square. So, putting all of that together, you get the expected value of x square minus expected value of x the whole square. So, this is another definition of the variance. But let us, this is just algebraic simplification.

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But let us try to understand what this function actually means. Variance of x is the expected value of x minus Ex the whole square. First thing you should ask is when is the variance 0. The variance is 0 when x minus Ex, x minus Ex the whole square is a positive quantity. When is the expectation of a positive quantity 0? Well, the expectation of positive quantity is 0 only when the entire thing always 0, which means that x minus Ex the whole square is 0, which is means that x is a constant. So, the only way the variance can be 0 if x is a constant, which is equal to the mean or the expectation.

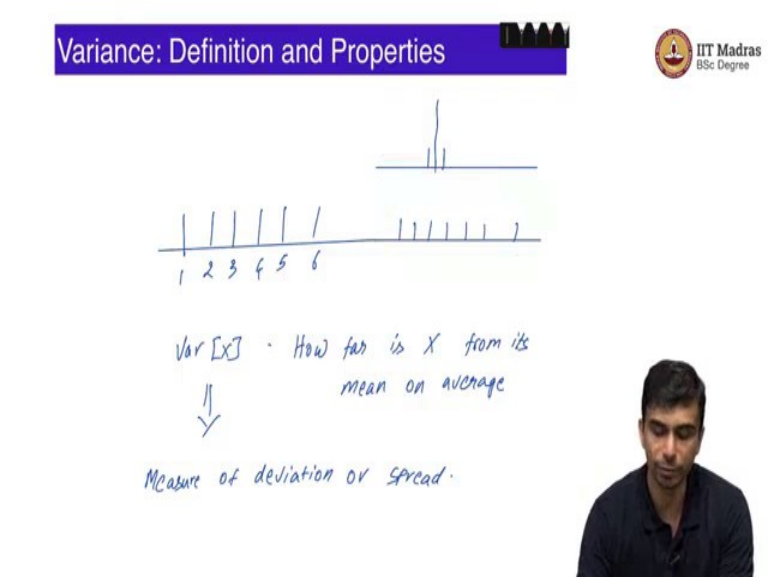
Let us compute the variance for some simple examples. Let us say x, let us we have the double example would be the double dice throw experiment. X is equal to, let us say, first dice face. What is the variance of x? So, let us, Ex we know already is equal to 3.5. What is the expected, now x minus Ex the whole square let us call y to be equal to x minus Ex the whole square. And what is the PMF of this particular random variable. So, let us probability of y is equal to, what all

values can this take? Well, it can take, x can take values 1, 2, 3, 4, 5, 6. So, when x is equal to 1, y, essentially that is just write in terms of x and y, x can take values 1, 2, 3, 4, 5, 6.

When x equal to 1, y is 1 minus 3.5 the whole squared which is 6.25, 2.5 whole square. When x is 2 this is 2 minus 3.5 whole square which is 1.5 the whole square which is 2.25. When x is equal to 3, it is 3 minus 3.5 the whole square, which is 0.5 the whole square, which is 0.25, because it is symmetric, now 4 would also be 0.25, 2.25 and 6.25. So, y is x minus Ex the whole square and when x is 1 or 6, it is actually y is 6.25. When x is 2 or 5, it is 2.25. And then x is 3 or 4, it is 0.25. Now, we can compute by the simple probability of x being the simple first dice face result. Each of these 1 to 6 occur with probability 1 by 6.

So, you have 1 by 6, the variance are, would be the expected value of y, which means that 1 by 6 times 6.25 plus 1 by 6 times 2.25 plus 1 by 6 times 0.25, and so on. So, Ey, which is equal to variance of x would be equal to 1 by 6 times 6.25 plus 1 by 6 times 2.25 plus 1 by 6 times 0.25. This would be repeating two times. We can just multiply that by 2. So, this is the variance of x. You can compute the value here. So, what does this actually mean? Now, we have the simple example. But what does this actually mean? Let me just move this to the example part here.

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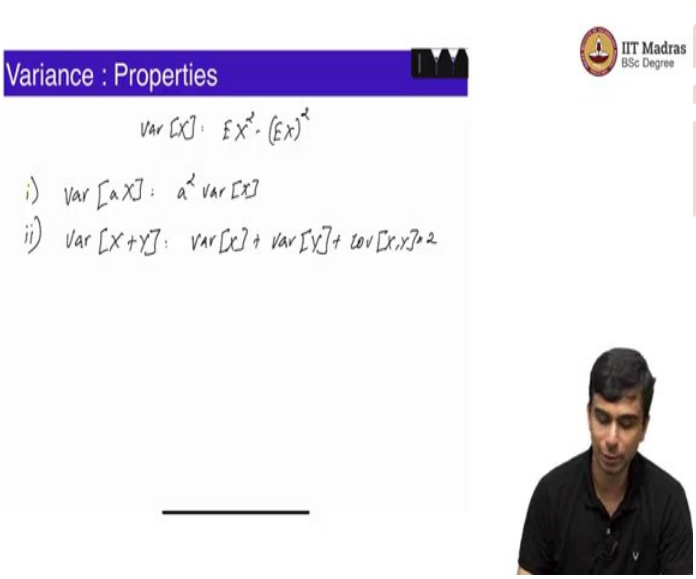
In this particular example, what we had was this. This was the probability mass function of the random variable x 1, 2, 3, 4, 5, 6. So, there is 1, 2, 3, 4, 5, 6. So, you have, this is the probability mass function of the random variable x, which was simply the result of the dice throw. X minus

Ex the whole square is 1 minus 3.5 the whole square plus 2 minus 3.5 the whole square, I mean, the average value of 1 minus, effectively it is measuring how far is x from its mean, so 1 minus 3.5 the whole square, the average of 1 minus 3.5 the whole square, 2 minus 3.5 the whole square, 3 minus 3.5 the whole square, 4 minus 3.5 the whole square, 5 minus 3.5 the whole square and 6 minus 3.5 the whole square.

So, this effectively is the square distance between x and its mean. So, variance of x essentially measures how far is x from its mean on average, on average how far is x from its mean. So, if x was actually a constant, then it would exactly be equal to its mean or expectation. I will be interchangeably using mean and expectation. So, if x was a constant, then x is always equal to its mean. So, it is at a distance of 0 from its mean. So, variance of x effectively captures how far is x from its mean on average. So, x, variance is set to capture, it is a measure of deviation or spread.

If you have a probability mass function, which is spread out from left to right, then the variance of x will be large. If you have a probability mass function which is quite concentrated, for example, you can imagine two different PMFs, let us say, PMF, let us have the same scale so that it is easy to compare, we can have a PMF like this, you can have another PMF like this. This particular PMF is less spread out when compared to this PMF. So, we can immediately say that the variance of this random variable is lesser than the variance of this random variable. So, the variance captures the measure of spread or deviation of the random variable from its mean.

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So, we will now take a look at some of the properties of the variance. So, we have defined the variance as expected value of x square minus Ex the whole square. Two of the main properties of the variance are as follows. First is variance of a times x is simply equal to a square times variance of x, where a is some constant, like for example, 2 or 4 or 75 or some number, some things are number.

And you can see that why we have a square unlike the case of expectation, where if you take a outside, you get a square here, in the case of expectation if you take a outside, you just get a. Expectation of a times x is a times expectation of x. But variance of ax is a square times variance of x.

And the other property is that it is actually requires knowledge of covariance, but I will define that right now and you go later. Variance of x plus y is equal to variance of x plus variance of y plus covariance of x comma y into 2. This can look, this looks familiar to lot of you in terms of a plus b the whole square is a square plus b square plus 2ab and in fact that is exactly the reason why you have this. But what is the covariance?

I will define the covariance later when we do multiple random variables. But the key thing to note is variance of x plus y is not equal to variance of x plus variance of y. So, variance of unlike the case of expectation where you had expected value of x plus y or simply Ex plus Ey, but variance of x plus y is not equal to variance of x plus variance of y.